

[AmebaPro2] Meta Analyzer Tool User Manual

2024.01.10



1. Tool Operation.

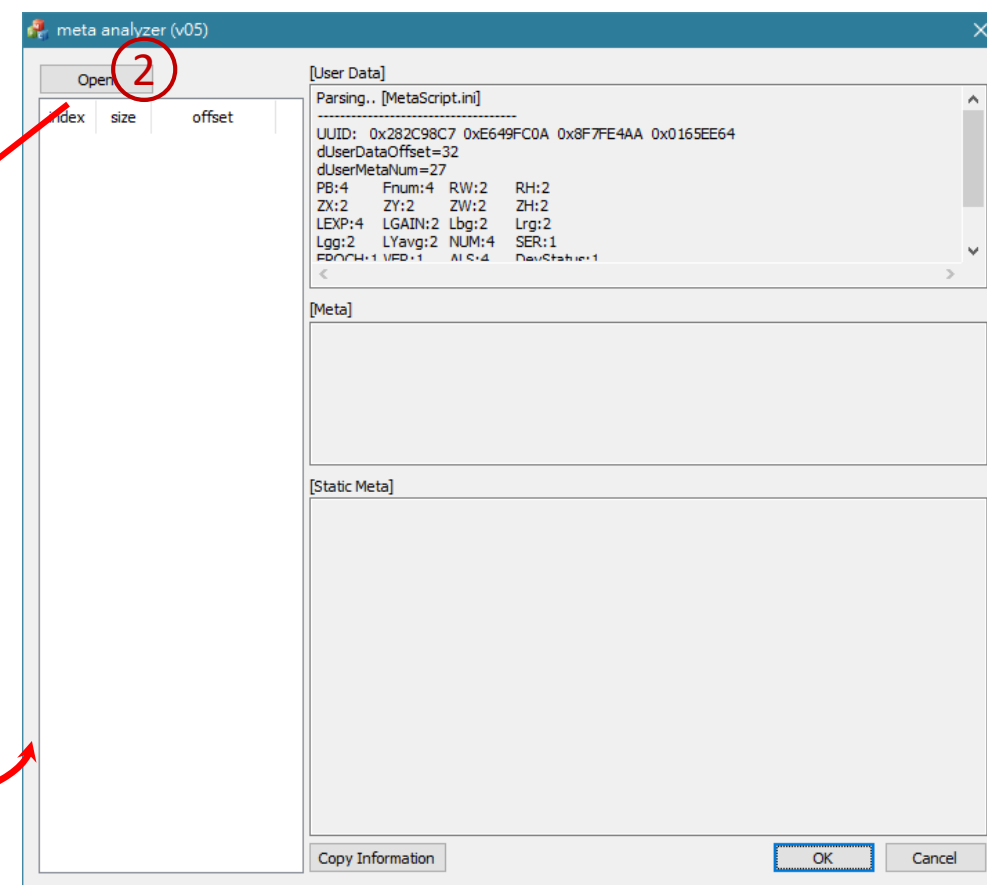
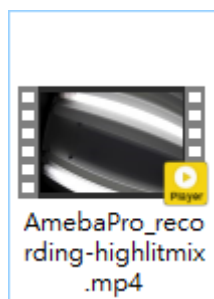
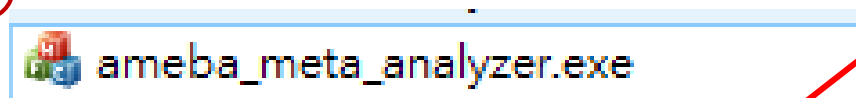
2. User Data and Script Introduction.

1. Execute "ameba_meta_analyzer.exe" to open the GUI
2. Auto generate the MetaScript.h in the same folder, which is used for coding in SDK.



3. Drag and drop the video to log window or click "Open" button to select video, meta index will be displayed in the left log window by each frame.

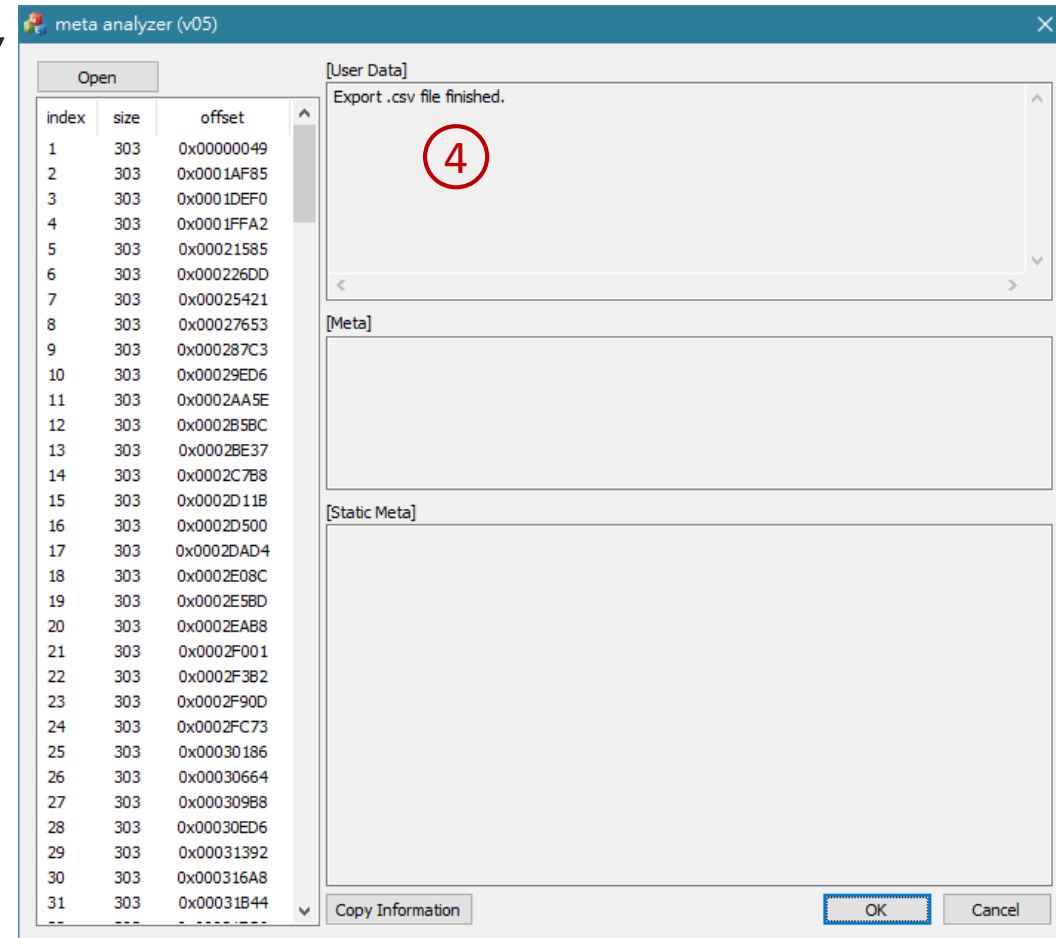
①



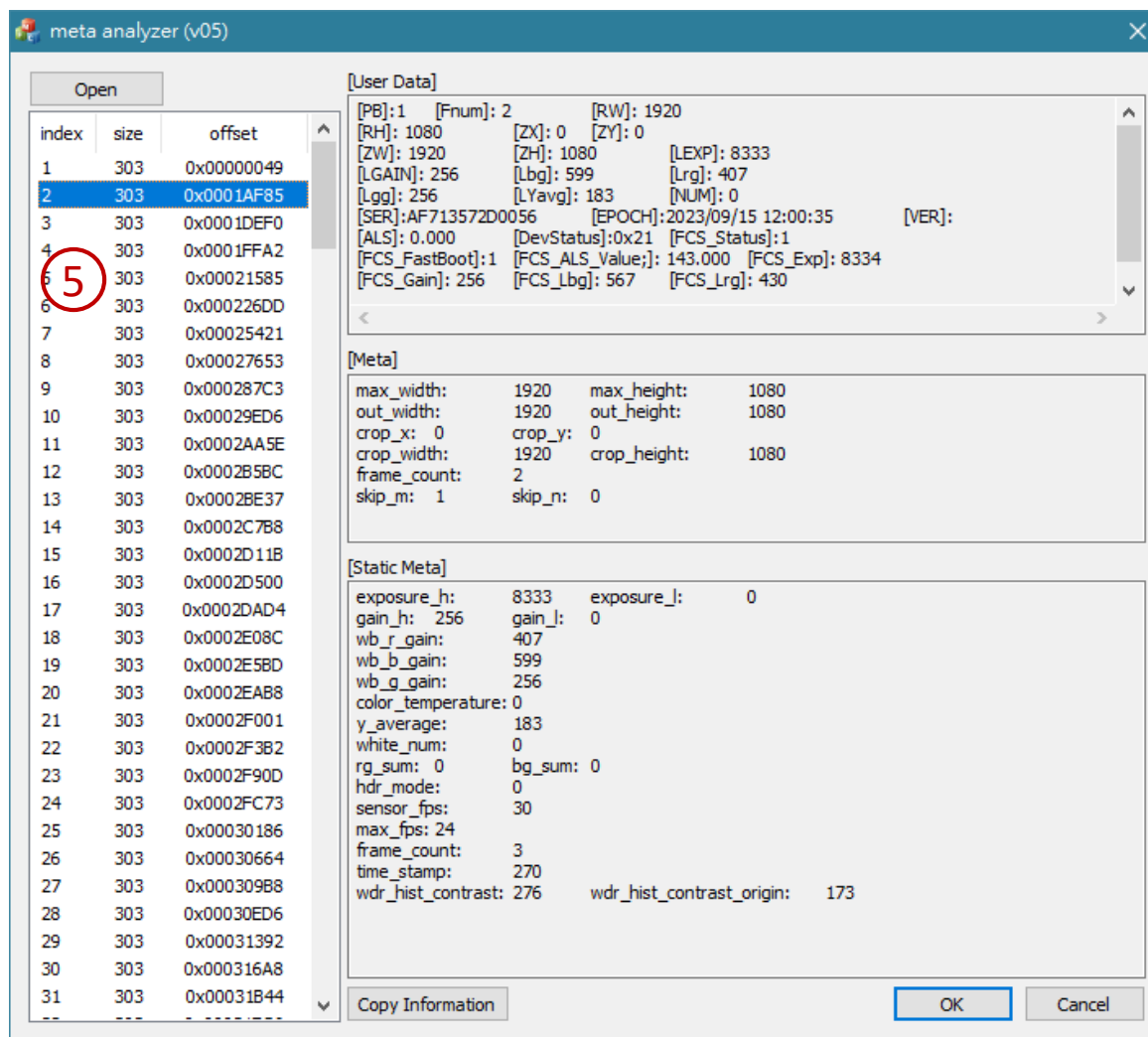
4. Export all the frame metadata to csv file in the same directory.
5. Generate .csv done when show message "Export .csv file finished"

③

	AmebaPro_recording-highlitmix.mp4	2023/9/15
	AmebaPro_recording-highlitmix.csv	2023/10/1



6. Click on the window on the left and it will show meta data to the right side.



7. Click on the index number with prefix [i], the right side expand the 3A information.

Note: Prefix [i] means i frame.

meta analyzer (Ver-09.01.03) - H:\AmebaPro_recording.mp4

Open [i] i-Frame

index	size	offset
268	303	0x0013759F
269	303	0x00137893
270	303	0x00137BA8
[i] 271	303	0x001380FF
272	303	0x00143A82
273	303	0x00143DA4
274	303	0x001440DE
275	303	0x00144634
276	303	0x0014491E
277	303	0x00144BD4
278	303	0x00144E91
279	303	0x0014539F
280	303	0x001456C4
281	303	0x00145A35
282	303	0x00145E2A
283	303	0x001464A0
284	303	0x00146B0E
285	303	0x0014728F
286	303	0x0014766F
287	303	0x00148DE9
288	303	0x0014910A
289	303	0x00149548
290	303	0x00149F20
291	303	0x0014A547
292	303	0x0014B081
293	303	0x0014B48C
294	303	0x0014BEBC
295	303	0x0014C351
296	303	0x0014DA86
297	303	0x0014DD89
298	303	0x0014E554
299	303	0x0014F508
300	303	0x0014F83C
[i] 301	303	0x0015040C
302	303	0x0016E6D5
303	303	0x0016F8A0

[User Data]

[PB]:0 [Fnum]: 270 [RW]: 1920
 [RH]: 1080 [ZX]: 0 [ZY]: 0
 [ZW]: 1536 [ZH]: 864 [LEXP]: 1666
 [LGAIN]: 360 [Lbg]: 662 [Lrg]: 424
 [Lgg]: 256 [LYavg]: 230 [NUM]: 716
 [SER]: RTK-TEST [EPOCH]: 2024/01/10 15:52:15 [VER]: 1.2.3
 [ALS]: 37.119 [DevStatus]: 0x21

[Meta]

max_width: 1920 max_height: 1080
 out_width: 1920 out_height: 1080
 crop_x: 0 crop_y: 0
 crop_width: 1920 crop_height: 1080
 frame_count: 271
 skip_m: 1 skip_n: 0

[Static Meta]

exposure_h: 10000 exposure_l: 0
 gain_h: 380(1.48) gain_l: 0(0.00) (ETGain: 148.4375)
 wb_r_gain: 354
 wb_b_gain: 565
 wb_g_gain: 256
 color_temperature: 4258
 y_average: 177
 white_num: 378
 rg_sum: 69978 bg_sum: 43820
 hdr_mode: 0
 sensor_fps: 30
 max_fps: 30
 frame_count: 271
 time_stamp: 14101
 wdr_hist_contrast: 0 wdr_hist_contrast_origin: 0

3A Info

[AE] frame_count:270 win_cnt:256
 703, 799, 1177, 2049, 3063, 3190, 4074, 4225, 3481, 2528, 2680, 1685, 1711, 1615, 1427, 1467, 1668, 1832, 1965, 1960, 1954, 1859, 1819, 1598, 1487, 1433, 1377, 1540, 1964, 1619, 1502, 1493, 1562, 1441, 1403, 1242, 1207, 1151, 1073, 984, 945, 879, 773, 669, 644, 679, 696, 623, 556, 509, 503, 480, 466, 469, 478, 429, 472, 411, 357, 348, 413, 371, 377, 400, 430, 448, 431, 502, 462, 471, 463, 469, 486, 445, 440, 460, 448, 457, 398, 406, 415, 386, 417, 395, 412, 436, 421, 433, 497, 443, 470, 479, 490, 471, 485, 495, 461, 525, 512, 494, 478, 441, 424, 368, 367, 357, 352, 348, 290, 271, 288, 312, 295, 277, 261, 185, 181, 201, 206, 217, 188, 179, 168, 158, 135, 157, 163, 170, 170, 142, 134, 144, 151, 123, 104, 084, 069, 092, 068, 090, 090, 093, 093, 086, 106, 099, 113, 096, 101, 119, 104, 092, 091, 099, 079, 067, 074, 071, 062, 081, 058, 051, 053, 056, 074, 060, 063, 054, 042, 055, 041, 061, 039, 034, 033, 031, 032, 027, 031, 027, 024, 023, 022, 029, 015, 017, 018, 012, 014, 017, 011, 016, 020, 011, 016, 010, 017, 009, 012, 017, 008, 012, 010, 013, 009, 012, 010, 004, 003, 009, 004, 005, 006, 004, 006, 004, 004, 005, 003, 010, 003, 006, 009, 006, 010, 008, 010, 008, 005, 005, 004, 005, 007, 006, 005, 000, 003, 005, 001, 004, 003, 007, 003, 004, 002, 003, 001, 004, 008, 205,

[AWB] frame_count:270 win_cnt:256
 (AWB-B/G)
 0.58, 0.57, 0.59, 0.58, 0.56, 0.51, 0.52, 0.54, 0.54, 0.54, 0.54, 0.53, 0.49, 0.30, 0.57, 0.57, 0.57, 0.57, 0.56, 0.49, 0.51, 0.53, 0.54, 0.54, 0.54, 0.54, 0.54, 0.48, 0.58, 0.56, 0.58, 0.57, 0.53, 0.49, 0.50, 0.48, 0.54, 0.54, 0.54, 0.54, 0.54, 0.54, 0.50, 0.55, 0.56, 0.56, 0.50, 0.48, 0.47, 0.43, 0.53, 0.54, 0.54, 0.54, 0.54, 0.54, 0.54, 0.42, 0.46, 0.49, 0.55, 0.46, 0.46, 0.44, 0.42, 0.52, 0.54, 0.54, 0.47, 0.52, 0.54, 0.54, 0.33, 0.41, 0.46, 0.45, 0.43, 0.41, 0.39, 0.42, 0.52, 0.52, 0.52, 0.41, 0.50, 0.54, 0.48, 0.33, 0.40, 0.45, 0.40, 0.40, 0.38, 0.38, 0.39, 0.51, 0.49, 0.45, 0.34, 0.39, 0.43, 0.41, 0.32, 0.41, 0.43, 0.41, 0.50, 0.61, 0.52, 0.38, 0.49, 0.37, 0.39, 0.33, 0.34, 0.35, 0.41, 0.34, 0.42, 0.44, 0.57, 0.67, 0.65, 0.52, 0.46, 0.43, 0.44, 0.39, 0.31, 0.31, 0.31, 0.41, 0.38, 0.42, 0.47, 0.62, 0.66, 0.64, 0.57, 0.47, 0.47, 0.43, 0.38, 0.31, 0.29, 0.33, 0.41, 0.39, 0.40, 0.34, 0.49, 0.64, 0.58, 0.50, 0.46, 0.47, 0.46, 0.32, 0.41, 0.27, 0.30, 0.21, 0.43, 0.47, 0.47, 0.46, 0.56, 0.50, 0.44, 0.45, 0.44, 0.44, 0.32, 0.29, 0.27, 0.37, 0.44, 0.49, 0.50, 0.51, 0.52, 0.52, 0.49, 0.44, 0.43, 0.45, 0.44, 0.38, 0.35, 0.36, 0.20, 0.44, 0.49, 0.50, 0.51, 0.52, 0.53, 0.54, 0.50, 0.28, 0.41, 0.41, 0.44, 0.44, 0.29, 0.25, 0.41, 0.48, 0.49, 0.50, 0.51, 0.52, 0.52, 0.40, 0.27, 0.29, 0.37, 0.36, 0.28, 0.27, 0.26, 0.21, 0.47, 0.48, 0.49, 0.50, 0.50, 0.49, 0.27, 0.27, 0.27, 0.27, 0.27, 0.27, 0.27, 0.27, 0.27, 0.21

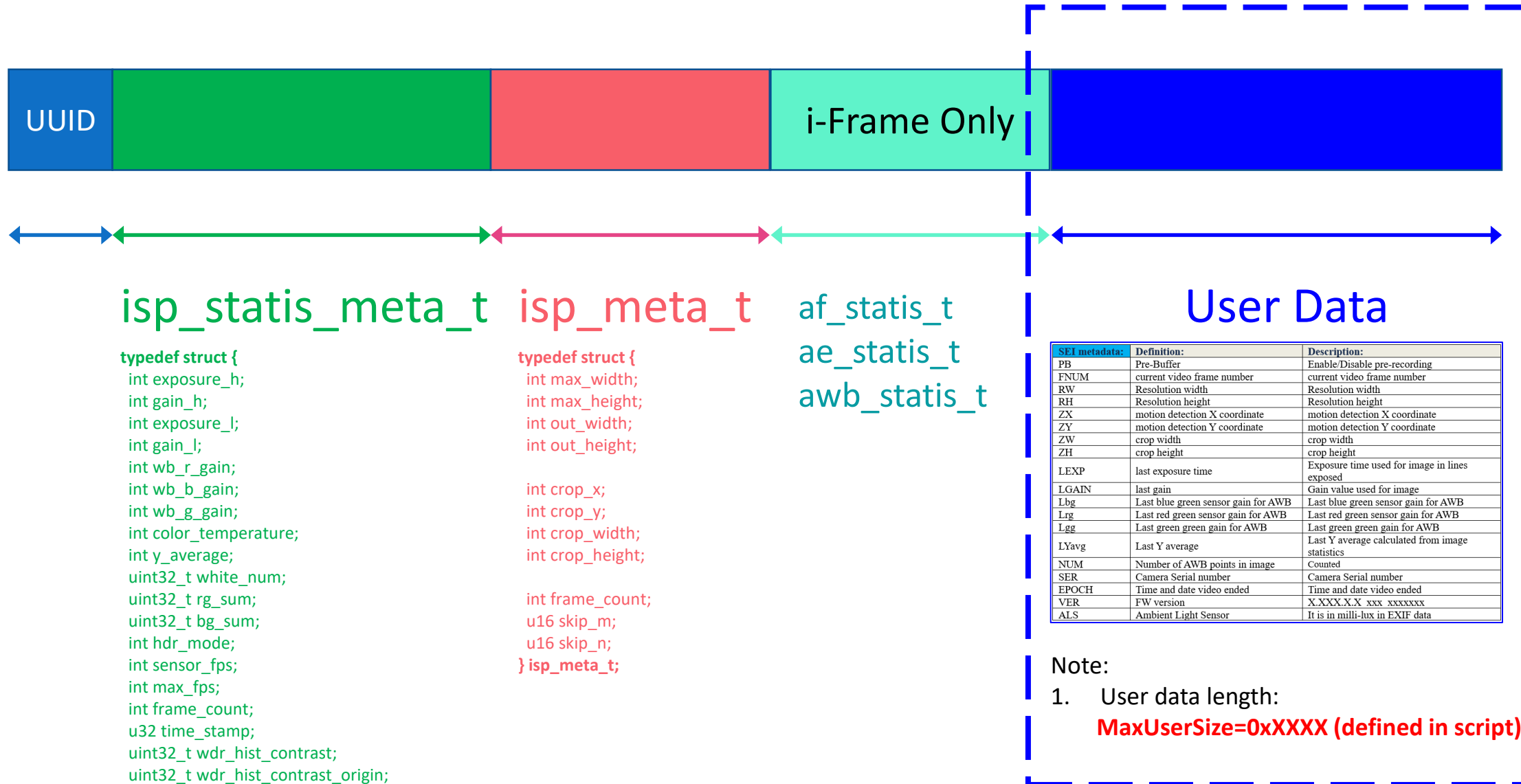
AE Histogram, AWB BG-map, AWB RG-map

OK

1. Tool Operation.

2. User Data and Script Introduction.

SEI Structure in Media File



User Data in *.mp4 Files

The table on the right is the basic items of user data in SEI.

In addition to the basic items, users can add or even change the content of basic items in **MetaScript.ini**, as long as they match the SEI structure in the SDK.

SEI metadata:	Definition:	Description:
PB	Pre-Buffer	Enable/Disable pre-recording
FNUM	current video frame number	current video frame number
RW	Resolution width	Resolution width
RH	Resolution height	Resolution height
ZX	motion detection X coordinate	motion detection X coordinate
ZY	motion detection Y coordinate	motion detection Y coordinate
ZW	crop width	crop width
ZH	crop height	crop height
LEXP	last exposure time	Exposure time used for image in lines exposed
LGAIN	last gain	Gain value used for image
Lbg	Last blue green sensor gain for AWB	Last blue green sensor gain for AWB
Lrg	Last red green sensor gain for AWB	Last red green sensor gain for AWB
Lgg	Last green green gain for AWB	Last green green gain for AWB
LYavg	Last Y average	Last Y average calculated from image statistics
NUM	Number of AWB points in image	Counted
SER	Camera Serial number	Camera Serial number
EPOCH	Time and date video ended	Time and date video ended
VER	FW version	X.XXX.X.X xxx xxxxxxxx
ALS	Ambient Light Sensor	It is in milli-lux in EXIF data

DevStatus

STS		Bitmask							
bit	8	7	6	5	4	3	2	1	0
de	CROSS_A	ALWAYS_ON_MODE	SIREN_ON_	AUTOHDR_ON	HDR_ON_	NIGHT_ON_	IRLED_ON_	SPOTLIGHT_O	IRCUT_ON_
sc	LERT	_ENABLED	STATE	_STATE	STATE	STATE	STATE	N_STATE	STATE

Table referenced source: “Arlo camera EXIF and Metdata specification.docx”

User Data - Script Files (.ini)

The script (MetaScript.ini) corresponding to the basic items.

[MetaData]

```

UUID={0xc7,0x98,0x2c,0x28,0x0a,0xfc,0x49,0xe6,0xaa,0xe4,0x7f,0x8f,0x64,0xee,0x65,0x01}
UserDataOffset=32
MaxUserSize=0x80
TotalParameter=20
PB=uint8[1] //prebuffer
Fnum=uint32[1] //current video frame number
RW=uint16[1] //Resolution width
RH=uint16[1] //Resolution height
ZX=uint16[1] //motion detection X coordinate
ZY=uint16[1] //motion detection Y coordinate
ZW=uint16[1] //crop width
ZH=uint16[1] //crop height
LEXP=uint32[1] //last exposure time
LGAIN=uint16[1] //last gain
Lbg=uint16[1] //Last blue sensor gain for AWB
Lrg=uint16[1] //Last red sensor gain for AWB
Lgg=uint16[1] //Last green sensor gain for AWB
LYavg=uint16[1] //Last Y average
NUM=uint32[1] //Number of AWB points in image
SER=uint8[16] //Camera Serial number
EPOCH=uint8[12] //Time and date video ended
VER=uint8[3] //FW version
ALS=uint32[1] //Ambient Light Sensor
DevStatus=uint8[1] //peripheral device status
    
```



SEI metadata:	Definition:	Description:
PB	Pre-Buffer	Enable/Disable pre-recording
FNUM	current video frame number	current video frame number
RW	Resolution width	Resolution width
RH	Resolution height	Resolution height
ZX	motion detection X coordinate	motion detection X coordinate
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LYavg	Last Y average	Last Y average calculated from image statistics
NUM	Number of AWB points in image	Counted
SER	Camera Serial number	Camera Serial number
EPOCH	Time and date video ended	Time and date video ended
VER	FW version	X.XXX.X.X xxx xxxxxxxx
ALS	Ambient Light Sensor	It is in milli-lux in EXIF data

STS		Bitmask							
bit	8	7	6	5	4	3	2	1	0
de	CROSS_A	ALWAYS_ON_MODE	SIREN_ON_	AUTOHDR_ON_	HDR_ON_	NIGHT_ON_	IRLED_ON_	SPOTLIGHT_O	IRCUT_ON_
sc	LERT	_ENABLED	STATE	_STATE	STATE	STATE	STATE	N_STATE	STATE

User Data - Definition

In script, the definition of the single item.

[MetaData]

```

UUID={0xc7,0x98,0x2c,0x28,0x0a,0xfc,0x49,0xe6,0xaa,0xe4,0x7f,0x8f,0x64,0xee,0x65,0x01}
UserDataOffset=32 ( Removed from tool version v09 )
MaxUserSize=0x80 ( Equal to the definition in firmware: #define VIDEO_META_USER_SIZE 0XXXXX )
TotalParameter=20 ( Must be larger than or equal to the number of all below parameters )
PB=uint8[1] //prebuffer
Fnum=uint32[1] //current video frame number
RW=uint16[1] //Resolution width
RH=uint16[1] //Resolution height
ZX=uint16[1] //motion detection X coordinate
ZY=uint16[1] //motion detection Y coordinate
ZW=uint16[1] //crop width
ZH=uint16[1] //crop height
LEXP=uint32[1] //last exposure time
LGAIN=uint16[1] //last gain
Lbg=uint16[1] //Last blue sensor gain for AWB
Lrg=uint16[1] //Last red sensor gain for AWB
Lgg=uint16[1] //Last green sensor gain for AWB
LYavg=uint16[1] //Last Y average
NUM=uint32[1] //Number of AWB points in image
SER=uint8[16] //Camera Serial number
EPOCH=uint8[12] //Time and date video ended
VER=uint8[3] //FW version
    
```

LEXP=uint32[1] //last exposure time

↔	↔	↔	↔
Name	Data Type	Number	Comment

User Data – Data Type

List of available data types

Data type	Print syntax	Display Range
uint8[]	"%d", (unsigned char)buf8[i]	0 ~ 255
uint16[]	"%d", (unsigned short)buf16[i]	0 ~ 65,535
uint32[]	"%d", (unsigned int)buf32[i]	0 ~ 4,294,967,295
int8[]	"%d", (char)buf8[i]	-128 ~ 127
int16[]	"%d", (short)buf16[i]	-32,768 ~ 32,767
int32[]	"%d", (int)buf32[i]	-2,147,483,648 ~ 2,147,483,647
char[]	"%s", (char*)buf8	ASCII with fixed length.
char*	"%s", (char*)buf8	ASCII and end with \0.

User Data – Display on Tool

SEI metadata:	Data type	Print syntax	Display	Comment
PB	uint8[1]	"%d", (unsigned char)buf8[i]	[PB]:1	
FNUM	uint32[1]	"%d", (unsigned int)buf32[i]	[Fnum]: 1105	
RW	uint16[1]	"%d", (unsigned short)buf16[i]	[RW]: 1920	
RH	uint16[1]	"%d", (unsigned short)buf16[i]	[RH]: 1080	
ZX	uint16[1]	"%d", (unsigned short)buf16[i]	[ZX]: 0	
ZY	uint16[1]	"%d", (unsigned short)buf16[i]	[ZY]: 0	
ZW	uint16[1]	"%d", (unsigned short)buf16[i]	[ZW]: 1920	
ZH	uint16[1]	"%d", (unsigned short)buf16[i]	[ZH]: 1080	
LEXP	uint32[1]	"%d", (unsigned int)buf32[i]	[LEXP]: 8333	
LGAIN	uint16[1]	"%d", (unsigned short)buf16[i]	[LGAIN]: 291	
Lbg	uint16[1]	"%d", (unsigned short)buf16[i]	[Lbg]: 627	
Lrg	uint16[1]	"%d", (unsigned short)buf16[i]	[Lrg]: 421	
Lgg	uint16[1]	"%d", (unsigned short)buf16[i]	[Lgg]: 256	
LYavg	uint16[1]	"%d", (unsigned short)buf16[i]	[LYavg]: 232	
NUM	uint32[1]	"%d", (unsigned int)buf32[i]	[NUM]: 1016	
SER	uint8/char[16]	""s"", char*	[SER]:'XXXXXX'	This item is forced to be displayed in printable ASCII code .
EPOCH	uint8[12]	"%04d/%02d/%02d %02d:%02d:%02d", uint16, uint16, uint16, uint16, uint16, uint16	[EPOCH]:2023/10/25 08:40:50	Store as uint8/16, and display as timestamp format.
VER	uint8/char[7]	""s"", char*	[VER]:'XX.OOO'	This item is forced to be displayed in printable ASCII code .
ALS	uint32[1]	" %.3f", (float)buf32[i]/1000	[ALS]: 297.201	Store as int/uint, but display by dividing 1000.
DevStatus	uint8[1]	"0x%02X", buf8[i]	[DevStatus]:0x00	This item is forced to be displayed in HEX .

User Data – Dynamic Message

User Data

SEI metadata:	Definition:	Description:
PB	Pre-Buffer	Enable/Disable pre-recording
FNUM	current video frame number	current video frame number
RW	Resolution width	Resolution width
RH	Resolution height	Resolution height
ZX	motion detection X coordinate	motion detection X coordinate
ZY	motion detection Y coordinate	motion detection Y coordinate
ZW	crop width	crop width
ZH	crop height	crop height
LEXP	last exposure time	Exposure time used for image in lines exposed
LGAIN	last gain	Gain value used for image
Lbg	Last blue green sensor gain for AWB	Last blue green sensor gain for AWB
Lrg	Last red green sensor gain for AWB	Last red green sensor gain for AWB
Lgg	Last green green gain for AWB	Last green green gain for AWB
LYavg	Last Y average	Last Y average calculated from image statistics
NUM	Number of AWB points in image	Counted
SER	Camera Serial number	Camera Serial number
EPOCH	Time and date video ended	Time and date video ended
VER	FW version	X.XXX.X.X xxx xxxxxxxx
ALS	Ambient Light Sensor	It is in milli-lux in EXIF data

Note:

- User data length:
MaxUserSize=0xFFFF (defined in script)



Start Identify Code
(0x8081)

End Identify Code
(0x8081)

"this_int1=13579" "this_string1=company"

"this_int2=13579" "this_string2=dynamic_SEI"

User Data
(Defined in MetaScript.ini)

SEI metadata:	Definition:	Description:
PB	Pre-Buffer	Enable/Disable pre-recording
FNUM	current video frame number	current video frame number
RW	Resolution width	Resolution width
RH	Resolution height	Resolution height
ZX	motion detection X coordinate	motion detection X coordinate
ZY	motion detection Y coordinate	motion detection Y coordinate
ZW	crop width	crop width
ZH	crop height	crop height
LEXP	last exposure time	Exposure time used for image in lines exposed
LGAIN	last gain	Gain value used for image
Lbg	Last blue green sensor gain for AWB	Last blue green sensor gain for AWB
Lrg	Last red green sensor gain for AWB	Last red green sensor gain for AWB
Lgg	Last green green gain for AWB	Last green green gain for AWB
LYavg	Last Y average	Last Y average calculated from image statistics
NUM	Number of AWB points in image	Counted
SER	Camera Serial number	Camera Serial number
EPOCH	Time and date video ended	Time and date video ended
VER	FW version	X.XXX.X.X xxx xxxxxxxx
ALS	Ambient Light Sensor	It is in milli-lux in EXIF data

User data length:
MaxUserSize=0xFFFF (defined in script)

Dynamic Message
(Not defined in MetaScript.ini)

Output Example:

```
'[msg]:this_int1=13579'
```

```
'[msg]:this_string1=company'
```

```
'[msg]:this_int2=13579'
```

```
'[msg]:this_string2=dynamic_SEI'
```

Note:

- Show by ASCII.
- In SEI, Each string end with '\0'

Thank You

CNBG 2023

